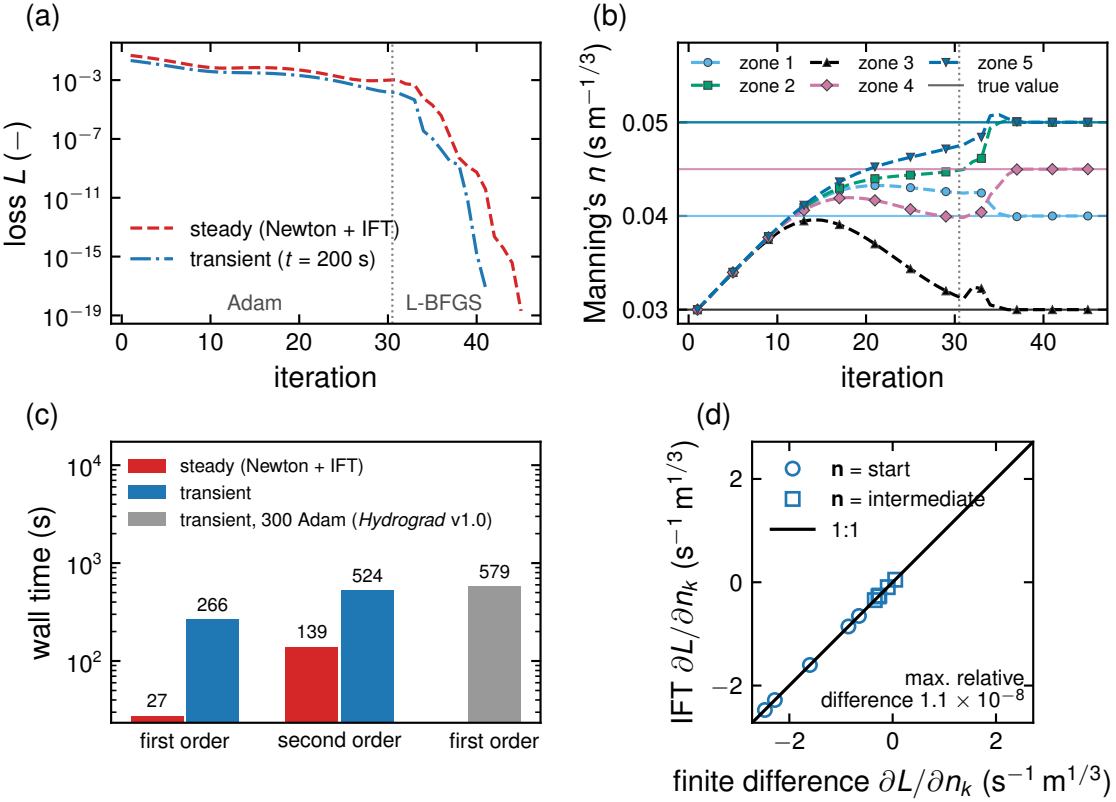

(a)
loss L (−)
10^{-3}
10^{-7}
10^{-11}
10^{-15}
10^{-19}
steady (Newton + IFT)
transient (t = 200 s)
Adam
L-BFGS
0
10
20
30
40
iteration
(b)
Manning's n (s m$^{-1/3}$)
zone 1
zone 2
zone 3
zone 4
zone 5
true value
0.05
0.04
0.03
0
10
20
30
40
iteration
(c)
wall time (s)
10^4
10^3
10^2
steady (Newton + IFT)
transient
transient, 300 Adam (*Hydrograd* v1.0)
27
266
139
524
579
first order
second order
first order
(d)
IFT $\partial L/\partial n_k$ (s^{-1} m$^{1/3}$)
n = start
n = intermediate
1:1
max. relative difference 1.1×10^{-8}
2
0
−2
−2
0
2
finite difference $\partial L/\partial n_k$ (s^{-1} m$^{1/3}$)